

# Silent Yet Expressive: Toward Seamless VR Communication through Emotion-aware Silent Speech Interfaces

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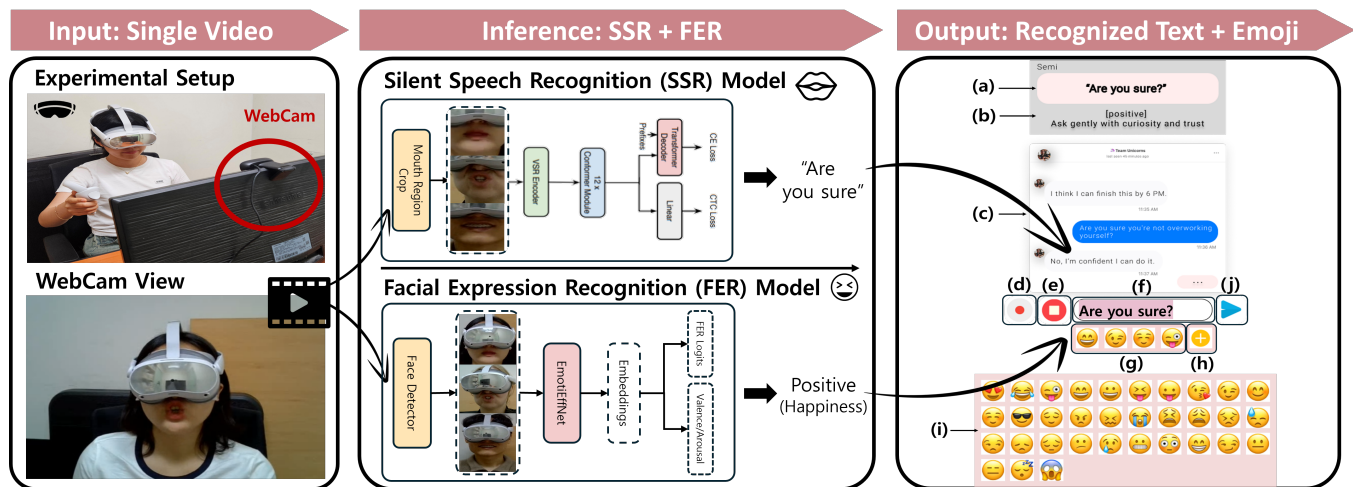
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**Figure 1: Overview of the emotion-aware silent-speech system pipeline.** Lip movement and facial expressions are captured via camera and processed for parallel silent-speech and emotion recognition, enabling expressive messaging in VR.

## Abstract

The increasing use of VR/MR in public spaces necessitates discreet and privacy-aware communication methods. While silent speech recognition offers an alternative to voice input, current systems cannot convey emotional nuance, limiting natural interaction. To bridge this gap, we present a novel emotion-aware silent-speech interface designed for immersive VR environments, which integrates facial expression recognition to enable the expression of affective cues. Our preliminary user study validates the feasibility of the proposed approach, showing promising results for enhanced, expressive communication within public VR contexts.

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## CCS Concepts

• Human-centered computing → Interaction techniques.

## Keywords

silent-speech, emotional expression, virtual reality

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## 1 Introduction

Speech recognition faces limitations in public settings due to privacy and social acceptance, making silent-speech recognition (SSR) a promising alternative [3, 7, 14, 18, 21]. SSR converts non-acoustic features—such as lip, tongue, or facial muscle movements—into text, thereby enabling users to engage in text-based communication without audible speech. This enables seamless interaction in

privacy/noise-sensitive or voice-restricted environments, providing greater social acceptability in public contexts [8, 15, 16]. The integration of SSR is particularly advantageous for the increasingly pervasive use of Virtual Reality (VR) and Mixed Reality (MR) head-mounted displays (HMDs) [1, 4–6]. As devices like Apple Vision Pro become common, users engage in diverse activities in public spaces: from attending virtual meetings in cafes to YouTube live streaming while traveling. In these everyday VR/MR contexts, SSR can offer unparalleled benefits for seamless and efficient communication [21]. However, a critical challenge in current SSR methods lies in their primary focus on converting articulatory movements into lexical content, often overlooking consideration for affective cues. Previous systems are predominantly optimized for phonetic and linguistic inference, resulting in an "expressiveness deficit" that significantly impedes the conveyance of emotional nuances essential for rich and socially meaningful interactions [2, 10]. Nevertheless, research on emotion recognition within silent speech contexts, particularly within VR/MR environments, remains extremely limited.

To address this gap, we present an emotion-aware silent-speech interface for seamless communication in VR environments. Specifically, we investigate the feasibility of integrating silent speech with facial expression recognition (FER) in immersive settings, specifically confronting challenges like partial facial occlusion by HMD. For this, we developed a working prototype and conducted a preliminary user study to evaluate the system's performance, usability, and user satisfaction within realistic VR messaging scenarios.

## 2 System Implementation

Our emotion-aware silent-speech interface for VR was developed as a messenger application for the PICO 4 Enterprise device using Unity 2021.3.13f1. Figure 1 shows the overall system pipeline, where a user's silent speech is translated into text and corresponding emojis. First, a Logitech 720p webcam captures the user's lip movements and facial expressions, with the video stream forwarded to the inference pipeline running on a single NVIDIA RTX 4090 GPU. This pipeline combines two distinct pretrained models in parallel to process a single video input: SSR for transcription and FER for capturing the user's emotions. For SSR, we utilized a visual-only AutoAVSR model [11], pretrained on the LRS3 dataset comprising 438.9 hours of TED talks. Mouth-region sequences, extracted via Mediapipe Face Mesh, are processed to generate transcribed text through a VSR encoder, Conformer blocks, and dual decoders with CTC and gloss-level objectives using Transformer Decoder. For FER, MTCNN detects and crops faces, which are then input to EmotiEffNet [17], a lightweight model trained on AffectNet dataset, to produce emotion logits across seven categories (anger, disgust, fear, happiness, sadness, surprise, neutral). These are mapped into three broader categories: Positive (Happiness), Negative (Anger, Disgust, Fear, Sadness), and Neutral (Neutral, Surprise). The system randomly selects 4 emojis matching the predicted emotion by leveraging a dataset of 74 emojis categorized as Positive, Negative, or Neutral [9]. Finally, the transcribed text and emotion-matched emojis are presented to the user via a VR messenger interface. Note that, in this work, no additional training or fine-tuning was made for both SSR and FER models.

## 3 User Study

The user study, conducted in a laboratory with 10 participants (4 males and 6 females;  $M = 25.6$ ,  $SD = 5.7$  years old), aimed to evaluate the model's accuracy and to assess the usability and user satisfaction of the proposed system under realistic constraints. After providing informed consent and demographic information, participants completed a brief training session using phrases different from those in the main task to become familiar with the silent speech and emotion recognition system. During the main task, participants were asked to send messages using silent speech with emotional expression. In each trial, participants first reviewed a phrase to communicate (Figure 1(a)) along with emotional expression guidelines (Figure 1(b)) within the dialogue context (Figure 1(c)). Participants then recorded their speech by pressing a button (recording: Figure 1(d), stopping: Figure 1(e)). The recorded video was processed to generate both transcribed text (Figure 1(f)) and a set of predicted emojis (Figure 1(g)). Participants could select from four suggested emojis or browse the full emoji set (Figure 1(h,i)) before sending the message (Figure 1(j)). This design follows a prior study that found a strong user preference for semi-automated emoji selection in VR interactions [12]. Each participant completed 30 trials (10 phrases  $\times$  3 emotions), resulting in a total of 300 trials. The phrases were drawn from the LRS3 dataset (English video corpus of TED talks) [11] and Enron datasets (Everyday email conversations) [20]. During each trial, the system automatically recorded the transcribed text, the predicted emotion label, and the inference time. After the experiment, participants completed 7-point scale post-questionnaires and an interview. Each session lasted approximately 50 minutes, and participants received an 8 USD compensation.

## 4 Results and Discussion

**SSR Performance:** The SSR model achieved a Character Error Rate of 22.25% and a Word Error Rate of 28.46%, approaching the performance levels of prior SSR studies [11, 14, 19, 21]. Importantly, P5 remarked that recognition quality exceeded expectations.

**FER Performance:** The FER model yielded a relatively low 3-class (Positive, Neutral, Negative) classification accuracy of 54.67%, falling below typical FER benchmarks [13]. This reduced performance might stem from HMD-induced facial occlusion and hardware constraints, with P4 reporting that the headset's weight hindered natural facial expression.

**Inference Time:** The average inference time was 7.1 seconds per trial (5.0s for SSR, 7.0s for FER), demonstrating the possibility of near real-time interaction. In particular, P1 noted the system was faster than expected, and P6 expressed interest in future use with improved responsiveness.

**Post-questionnaire:** Participants reported moderate to high overall satisfaction. They found the transcribed text generally aligned with their intent ( $M = 5.3$ ,  $SD = 1.0$ ,  $N = 7$ ) and gave moderate ratings to emoji recommendations ( $M = 4.9$ ,  $SD = 1.3$ ) while valuing its expressive potential. The emoji set design, rather than direct emotion mapping, likely compensated for FER limitations. In addition, participants perceived high usefulness ( $M = 5.7$ ,  $SD = 1.6$ ) and indicated strong willingness for future use ( $M = 5.6$ ,  $SD = 1.3$ ), particularly emphasizing hands-free, silent communication in voice-restricted settings and noting the novelty of emoji integration.

In summary, despite various technical challenges, these results suggest that integrating silent speech and emotion recognition offers a promising approach for expressive, voice-free communication in immersive VR environments.

## 5 Conclusion and Future Work

In this paper, we introduced an emotion-aware silent-speech interface that enables users to communicate both linguistic content and emotional intent in VR. Through a user study, we demonstrated the feasibility of combining SSR and FER within an immersive interface. As a preliminary investigation, this work highlights key directions for future development. First, the external webcam setup caused occlusion-related performance drops and lacks practicality for real-world use; built-in facial tracking or compact/integrated cameras may enhance accuracy and portability. Second, faster inference is needed for real-time use, suggesting optimized pipelines or lightweight models. Finally, future validation in real-world MR and public settings is needed to assess broader applicability.

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